

Technology #155

replace conventional aluminum oxide grinding
wheel with green silicon carbide wheel for
titanium alloy grinding

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-08-31

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding

WHY DOES IT MATTER?

Documented manufacturing problem: when grinding titanium alloys (Ti-6Al-4V) with conventional aluminum-oxide wheels, titanium chips adhere to the wheel's abrasive grains and bond (wheel loading); the loaded wheel stops cutting, friction rises, the wheel glazes, and the workpiece surface temperature spikes — producing tensile residual stresses, alpha-case burn, microcracking, and scrapped expensive aerospace parts. Titanium's low thermal conductivity and chemical reactivity with abrasive make this worse than steel grinding. Qualitative engineering-literature context; no fabricated statistics (retrieval must custody the evidence).

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in thermal. 9-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 9-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: comparative grinding test measuring wheel loading rate, workpiece temperature, surface roughness, and residual stress between conventiona.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: temperature-rise mapping at max duty. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden; interface contact resistances dominate at small scale and must be measured, not assumed zero; steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k; one-dimensional gradient (lateral gradients neglected); TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.; TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.; TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.; Model input variables requiring sourced values before any numeric result: h (W/(m².K)), A (m²), dT (K), L_i (m), k_i (W/(m.K)), A_i (m²), k (W/(m.K)), A (m²), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , W/(m.K)] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m²] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: green silicon carbide wheel maintained better thermal stability and wear resistance

POTENTIAL VALUE DRIVER: Addresses: Documented manufacturing problem: when grinding titanium alloys (Ti-6Al-4V) with conventional aluminum-oxide wheels, titanium chips adhere to the wheel's abrasive grains and bond (wheel loading); the loaded wheel stops cutting, friction rises, the wheel glazes, and the workpiece surface temperature spikes — producing tensile residual stresses, alpha-case burn, microcracking, and scrapped expensive aerospace parts. Titanium's low thermal conductivity and chemical reactivity with abrasive make this worse than steel grinding. Qualitative engineering-literature context; no fabricated statistics (retrieval must custody the evidence).

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: thermal

Disciplines: heat transfer, thermal engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding

Subsystems:

?: heat source characterization - ?

?: conduction path / spreader - ?

?: reject interface - ?

?: safety limit interlock - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding

Key Physics: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: CONDITIONAL; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: NOT ESTABLISHED.

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected). Engaged variables: k .

Assumptions:

- h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden
- interface contact resistances dominate at small scale and must be measured, not assumed zero
- steady-state conduction (no transient term)
- isotropic, homogeneous medium conductivity k
- one-dimensional gradient (lateral gradients neglected)
- TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.
- TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.
- TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.

- Model input variables requiring sourced values before any numeric result: h (W/(m².K)), A (m²), dT (K), L_i (m), k_i (W/(m.K)), A_i (m²), k (W/(m.K)), A (m²), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , W/(m.K)] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m²] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'wheel loading and glazing causing thermal damage to the workpiece', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'hotspot exceeding material/safety limit', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'interface delamination raising resistance over life', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coolant/flow path blockage (if active)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'thermal runaway coupling to electronics', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'condensation at cold surfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	The intervention must not require a different machine tool, must keep wheel	SOURCE_FACT
DI-002	Mechanism driver (proposed)	green silicon carbide wheel maintained better thermal stability and wear resis	MODELLED
DI-003	Expected effect target	reduced wheel loading, lower heat generation, maintained cutting ability, prev	MODELLED
DI-004	Domain design input pattern: distribution applicability by SYSTEM ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: ambient and applying by INDUSTRY ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: safety criterion applicability by [ELECTROSTATIC DISCHARGE] [temperature]	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance with standards (standards) [ELECTROSTATIC DISCHARGE] [temperature]	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-008	Custodied observation: Application of heat source for grinding of titanium alloy	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-009	Custodied observation: Exploring the effect of heat source on grinding of titanium alloy	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-010	Custodied observation: Mathematical modeling of grinding of titanium alloy	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-011	Custodied observation: Machine learning-based prediction of grinding of titanium alloy	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-012	Custodied observation: Industrial application of grinding of titanium alloy	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	heat source characterization realizing replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding	CONCEPTUAL	heat source characterization realizing replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding
DO-002	conduction path / spreader realizing replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding	CONCEPTUAL	conduction path / spreader realizing replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding
DO-003	reject interface realizing replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding	CONCEPTUAL	reject interface realizing replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding
DO-004	safety limit interlock realizing replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding	CONCEPTUAL	safety limit interlock realizing replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding
DO-005	heat source characterization interfaces with conduction path / spreader: signal energy transfer (for dimension, tolerance)	CONCEPTUAL	heat source characterization interfaces with conduction path / spreader: signal energy transfer (for dimension, tolerance)
DO-006	conduction path / spreader interfaces with reject interface: signal energy transfer (for dimension, tolerance)	CONCEPTUAL	conduction path / spreader interfaces with reject interface: signal energy transfer (for dimension, tolerance)
DO-007	reject interface interfaces with safety limit interlock: signal energy transfer (for dimension, tolerance)	CONCEPTUAL	reject interface interfaces with safety limit interlock: signal energy transfer (for dimension, tolerance)
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data processing (for dimension, tolerance)	CONCEPTUAL	Device boundaries: patient/fluid inlet and outlet terminal control/data processing (for dimension, tolerance)
DO-009	Flow/force/signal path geometry realizing the mechanism holds its state (for dimension, tolerance)	CONCEPTUAL	Flow/force/signal path geometry realizing the mechanism holds its state (for dimension, tolerance)
DO-010	Operating range for heat generation at source within which the mechanism holds its state (for dimension, tolerance)	CONCEPTUAL	Operating range for heat generation at source within which the mechanism holds its state (for dimension, tolerance)
DO-011	Operating range for spreader conductivity within which the mechanism holds its state (for dimension, tolerance)	CONCEPTUAL	Operating range for spreader conductivity within which the mechanism holds its state (for dimension, tolerance)
DO-012	Operating range for interface contact resistance within which the mechanism holds its state (for dimension, tolerance)	CONCEPTUAL	Operating range for interface contact resistance within which the mechanism holds its state (for dimension, tolerance)
DO-013	Operating range for reject-path area within which the mechanism holds its state (for dimension, tolerance)	CONCEPTUAL	Operating range for reject-path area within which the mechanism holds its state (for dimension, tolerance)
DO-014	Operating range for allowable temperature rise within which the mechanism holds its state (for dimension, tolerance)	CONCEPTUAL	Operating range for allowable temperature rise within which the mechanism holds its state (for dimension, tolerance)
DO-015	Operating range for ambient worst case within which the mechanism holds its state (for dimension, tolerance)	CONCEPTUAL	Operating range for ambient worst case within which the mechanism holds its state (for dimension, tolerance)
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism holds its state (for dimension, tolerance)	CONCEPTUAL	Sensing -> decision -> actuation loop maintaining the mechanism holds its state (for dimension, tolerance)
DO-017	Information path: sensing element -> decision/monitor control/data processing (for dimension, tolerance)	CONCEPTUAL	Information path: sensing element -> decision/monitor control/data processing (for dimension, tolerance)
DO-018	Fixture enabling: temperature-rise mapping at max duty (for dimension, tolerance)	CONCEPTUAL	Fixture enabling: temperature-rise mapping at max duty (for dimension, tolerance)
DO-019	Fixture enabling: interface resistance characterization (for dimension, tolerance)	CONCEPTUAL	Fixture enabling: interface resistance characterization (for dimension, tolerance)
DO-020	Fixture enabling: worst-case ambient soak test — rig control (for dimension, tolerance)	CONCEPTUAL	Fixture enabling: worst-case ambient soak test — rig control (for dimension, tolerance)

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: temperature-rise mapping at maximum	UNKNOWN	Unknown concept name; fixture specific instrumentation selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
heat generation at source	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
spreader conductivity	UNKNOWN (no sourced value)	W/(m.K)	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
interface contact resistance	UNKNOWN (no sourced value)	K/W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
reject-path area	UNKNOWN (no sourced value)	m^2	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
allowable temperature rise	UNKNOWN (no sourced value)	K	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
ambient worst case	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
wheel loading and glazing	Failure by single head damage due to high impact	Design avoidance of failure	STABSYS-052 (equipment design review)	UNKNOWN - that tested as failure, mitigated
hotspot exceeding material safety limit	Failure by hotspot exceeding material safety limit	spread safety margin; heat-seal gas spreading; protect against high temperature	Equipment design review	UNKNOWN - high temperature condition hotspot modes (AST)
interface delamination	Failure by interface delamination	Minimize interface resistance; control of feed preparation	Equipment design review	UNKNOWN - gaps and defects candidate failure modes (AST)
coolant/flow path blockage	Failure by active/flow path blockage (if active)	particulate removal; backflushing	Equipment design review	UNKNOWN - choked or clogged failure modes (AST)
thermal runaway coupling	Failure by thermal runaway coupling	design policy; thermal analysis; raised leakage	Equipment design review	UNKNOWN - which raise candidate failure mode (AST)
condensation at cold surfaces	Failure by condensation at cold surfaces	operating at low dew point; low point cooling	Equipment design review	UNKNOWN - raised design candidate failure mode (AST)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
wheel loading and glazing	Faults by wheel loading and glazing to the point of design control	NOT ESTABLISHED	Power Overhaul work is scheduled at intervals of 1 year (1Yr) & 20kWh
hotspot exceeding material safety limit	Hotspot exceeding material/safety limit	Under gear heat generation on speed high side of wheel and speed limit	Hotspot limit (Abs X&S)
interface delamination raising risk of structural failure	Interface delamination raising risk of structural failure	ES select real life cycle expansion joint	ES select real life cycle expansion joint
coolant/flow path blockage (if active)	Coolant/flow path blockage (if active)	Passive flow path blockage (if active)	Passive flow path blockage (if active)
thermal runaway coupling	Thermal runaway coupling	Electrical runaway coupling	Electrical runaway coupling
condensation at cold surfaces	Condensation at cold surfaces	Condensation at cold surfaces	Condensation at cold surfaces

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	comparative grinding test measuring wheel loading	comparative grinding test measuring wheel loading	comparative grinding test measuring wheel loading	comparative grinding test measuring wheel loading
VF-002	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test
VF-003	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty
VF-004	temperature-rise mapping at maximum	temperature-rise mapping at maximum	temperature-rise mapping at maximum	temperature-rise mapping at maximum
VF-005	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration
VF-006	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty
VF-007	worst-case ambient soak with staged power	worst-case ambient soak with staged power	worst-case ambient soak with staged power	worst-case ambient soak with staged power

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	biocompatible thermal spreader (graphite/AlN)	graphite/AlN	electronics thermal practice	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
biocompatible thermal spreader	biocompatible thermal spreader	1	STANDARD	biocompatible thermal spreader	Not Established	STANDARD	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- bonded spreader + encapsulation: domain registry manufacturing route - candidate only; no process qualification exists
- thermal interface material application control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter heat generation at source [q W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter spreader conductivity [k_i W/(m.K)]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter interface contact resistance [R_total K/W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter reject-path area [A m^2]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter allowable temperature rise [dT K]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter ambient worst case [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (BSE-HSE-MS-7-failure mode analysis) (to be quoted)	physical verification (BSE-HSE-MS-7-failure mode analysis) (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope	per method	constraint-compliance envelope	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	comparative grinding	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	thermal-resistance trend	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	worst-case ambient	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 9-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.